

Exposure to beta-carbolines norharman and harman.

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The aromatic beta-carbolines norharman and harman have been implicated in a number of human diseases including Parkinson's disease, tremor, addiction and cancer. It has been shown that these compounds are normal body constituents formed endogenously but external sources have been identified. Here, we summarise literature data on levels of norharman and harman in fried meat and fish, meat extracts, alcoholic drinks, and coffee brews. Other sources include edible and medicinal plants but tobacco smoke has been identified as a major source. Exposure levels from these different dietary sources are estimated to a maximum of 4 microg norharman per kg body weight (bw) per day and 1 microg harman per kg bw per day. Exposure via tobacco smoke depends on smoking habits and type of cigarettes but can be estimated to 1.1 microg/kg bw for norharman and 0.6 microg/kg bw for harman per package of cigarettes smoked. Studies on toxicokinetics indicate that inhalative exposure leads to a rapid increase in plasma levels and high bioavailability of norharman and harman. Oral bioavailability is lower but there are indications that sublingual absorption may increase dietary uptake of beta-carbolines. Endogenous formation can be estimated to be 50-100 ng/kg bw per day for norharman and about 20 ng/kg bw per day for harman but these rates may increase with high intake of precursors. Biomarker studies on plasma levels of beta-carbolines reported on elevated levels of norharman, harman or both in diseased patients, alcoholics and following tobacco smoking or consumption of beta-carboline-containing food. Cigarette smoking has been identified as major influence but dietary exposure may contribute to exposure.